FOOTSTEP POWER GENERATION AND CROWD MONITORING WITH IOT ALERTS

Introduction:

This project demonstrates energy harvesting using piezoelectric sensors and real-time crowd monitoring using ESP32 and Blynk IoT.

Objectives:

- Harvest energy from footsteps
- Detect footsteps accurately
- Monitor crowd activity
- Generate alerts during crowd surge

Components Used:

ESP32, Piezo Discs, Bridge Rectifier, Capacitor, LED, Resistors, Blynk IoT

Working Principle:

Piezoelectric sensors generate voltage when stepped on. The signal is processed by ESP32 and sent to Blynk for monitoring.

Results:

The system successfully detects steps, sends live data, and triggers alerts.

Conclusion:

The project proves the feasibility of using piezoelectric sensors and IoT for crowd safety monitoring.